WORKED SOLUTION

STEP I 1990, Question 4

Pure • Arithmetic Progressions and Differences

STEP I 1990, Question 4

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/90-s1-q4

The key idea

Every term of a progression is the first term plus a running total of differences. For an arithmetic progression the differences are constant, so the sum is easy. For a double arithmetic progression the differences themselves form an arithmetic progression, so we get one more layer of summing, and the binomial coefficients $\binom{n}{2}$ and $\binom{n}{3}$ appear naturally. The last part then becomes two linear equations for the two unknown differences.

Solution

Arithmetic progressions

Write $d = a_2 - a_1$ for the common difference, so $a_k = a_1 + (k-1)d$. Then

$$\sum_{k=1}^{n} a_k = na_1 + d\sum_{k=1}^{n}(k-1) = na_1 + \frac{n(n-1)}{2}(a_2 - a_1),$$

using $0 + 1 + \cdots + (n-1) = \frac{1}{2}n(n-1)$. Collecting the terms in a_1 and a_2,

$$a_1 + a_2 + \cdots + a_n = \frac{n}{2}[(3-n)a_1 + (n-1)a_2].$$

Double arithmetic progressions

Let the differences be $\delta_k = b_{k+1} - b_k$. These form an arithmetic progression with first term $p = \delta_1 = b_2 - b_1$ and common difference

$$e = \delta_2 - \delta_1 = (b_3 - b_2) - (b_2 - b_1) = b_3 - 2b_2 + b_1.$$

So $\delta_j = p + (j-1)e$, and adding up the first $k-1$ differences (the formula from the first part, with n replaced by $k-1$) gives

$$b_k = b_1 + \sum_{j=1}^{k-1}\delta_j = b_1 + (k-1)p + \frac{(k-1)(k-2)}{2}e = b_1 + \binom{k-1}{1}p + \binom{k-1}{2}e.$$

Now sum over $k = 1, \ldots, n$. We need two sums. The first is $\sum_{k=1}^{n}(k-1) = \binom{n}{2}$, as before. For the second, with $j = k-1$ running from 0 to $n-1$,

$$\sum_{j=0}^{n-1} \frac{j(j-1)}{2} = \frac{1}{2}\left[\frac{(n-1)n(2n-1)}{6} - \frac{(n-1)n}{2}\right] = \frac{n(n-1)}{12}[(2n-1)-3]$$
$$= \frac{n(n-1)(n-2)}{6} = \binom{n}{3}.$$

Therefore

$$\sum_{k=1}^{n} b_k = nb_1 + \binom{n}{2}(b_2 - b_1) + \binom{n}{3}(b_3 - 2b_2 + b_1).$$

This is already a formula in n, b_1, b_2, b_3. If you prefer the coefficient of each b_i separately, expanding gives

$$b_1 + \cdots + b_n = \frac{n(n^2 - 6n + 11)}{6}\, b_1 + \frac{n(n-1)(7-2n)}{6}\, b_2 + \frac{n(n-1)(n-2)}{6}\, b_3.$$

(Quick test: $n = 1, 2, 3$ give b_1, $b_1 + b_2$ and $b_1 + b_2 + b_3$.)

? **Ask yourself** Is there a neater way to see where $\binom{n}{3}$ comes from?

Yes: $\binom{k-1}{2}$ is the number of ways to choose 2 things from $k-1$, and the hockey-stick identity $\sum_{k=1}^{n}\binom{k-1}{2} = \binom{n}{3}$ (choose 3 from n by first fixing the largest, k) does the summing in one step. Each layer of "differences of differences" raises the binomial coefficient by one, just as integrating raises a power by one.

The factorial progression

Now $b_1 = 1$, and with p and e as above, $b_k = 1 + (k-1)p + \frac{1}{2}(k-1)(k-2)e$. So

$$b_2 = 1 + p, \qquad b_4 = 1 + 3p + 3e, \qquad b_6 = 1 + 5p + 10e.$$

In a factorial progression with $c_1 = b_2$, $c_2 = b_4$, $c_3 = b_6$, $c_4 = 220$, the successive differences are $1!\,d$, $2!\,d$, $3!\,d$, that is, d, $2d$, $6d$:

$$b_4 - b_2 = 2p + 3e = d,$$
$$b_6 - b_4 = 2p + 7e = 2d,$$
$$220 - b_6 = 219 - 5p - 10e = 6d.$$

Substituting the first into the second, $2p + 7e = 4p + 6e$, so $e = 2p$ and then $d = 8p$. The third equation becomes

$$219 - 5p - 20p = 48p \quad \Longrightarrow \quad 73p = 219 \quad \Longrightarrow \quad p = 3,$$

so $e = 6$ and $d = 24$, which is non-zero as required. The solution is unique, since every step was forced.

So $b_k = 1 + 3(k-1) + 3(k-1)(k-2) = 3k^2 - 6k + 4$, giving the sequence $1, 4, 13, 28, 49, 76, \ldots$, and indeed $b_2, b_4, b_6, 220 = 4, 28, 76, 220$ have differences $24, 48, 144 = 1! \cdot 24,\ 2! \cdot 24,\ 3! \cdot 24$.

Finally, using the sum formula with $b_1 = 1$, $b_2 - b_1 = 3$ and $b_3 - 2b_2 + b_1 = 6$:

$$\begin{aligned} 1 + b_2 + \cdots + b_n &= n + 3 \cdot \frac{n(n-1)}{2} + 6 \cdot \frac{n(n-1)(n-2)}{6} \\ &= \frac{n}{2}[2 + 3(n-1) + 2(n-1)(n-2)] \\ &= \frac{n(2n^2 - 3n + 3)}{2}. \end{aligned}$$

Checks

For $n = 1, 2, 3$ the final formula gives 1, 5 and 18, matching 1, $1 + 4$ and $1 + 4 + 13$. For $n = 6$ it gives $3 \cdot 57 = 171$, and directly $1 + 4 + 13 + 28 + 49 + 76 = 171$. A computer also checked both general sum formulas exactly (with fractions) for hundreds of random progressions and all $n \leq 24$.

Common slips

- Taking the differences of the factorial progression as $d, 2d, 3d$: they are $1!\,d$, $2!\,d$, $3!\,d$, so the third is $6d$.
- Writing $b_k = b_1 + kp + \ldots$: there are only $k - 1$ differences between b_1 and b_k.

Beyond STEP

This is the calculus of finite differences: a sequence whose mth differences are constant is a polynomial of degree m in n, and its partial sums are a polynomial of degree $m + 1$, best written in the basis $\binom{n}{0}, \binom{n}{1}, \binom{n}{2}, \ldots$ (Newton's forward-difference formula). Summing $\binom{n}{r}$ to get $\binom{n+1}{r+1}$ is the discrete twin of $\int x^r \, \mathrm{d}x = \frac{x^{r+1}}{r+1}$. For more in this spirit, try the problems in Siklos, *Advanced Problems in Mathematics*.

Beyond STEP worked solutions: https://beyondstep.keithmcnulty.org/. Solution text CC0 1.0; the question itself is © Cambridge University Press & Assessment and is not reproduced here.